Case study: Simulation

Lauri Paloneva	Marek Nečada	Khanh Ha	Hang Phung
908445	511832	1012042	101016692

January 12, 2026

1 Executive summary

Results

Oven	Mean (€)	Std (€)	Min (€)	Max (€)
Oven 0	−818707	35142	−962009	−693637
Oven 1	−2664647	1304917	−11897537	−784279
Oven 2	−504079	43411	−743706	−367686
Oven 3	−260137	23101	−352228	−187663

Table 1: NPVs for oven investments

Oven 3 offers the best value over a 5 year investment period based on the NPV Simulation results. *table 1 and figure 1a-d*

The storage costs explode for Oven 1 due to limited capacity (*Table 5*, Figure 1b and 2b). The cumulative storage grows large and constitutes a huge cost for the total NPV.

Energy costs and savings constitute a large portion of the total NPV Energy costs are on average around 500k€ over the investment horizon, while repair costs are around 160k€, and storage costs range from 30k€ to 120k€ (excluding Oven 1 with huge costs) (*Table 5 and figure 1a-d*). Investing in energy efficiency seems to provide the most value.

Large Repair and energy costs, with limited energy savings cause the Current oven to be more expensive than options 2 and 3. Additional storage compared to these options does not justify the risk of catastrofic failure or the really high energy costs (*Table 5*)

The Energy savings by themselves from Oven 3 justifies the higher investment and maintenance fees compared to the other ovens. The energy savings for oven 3 are 240k€ higher compared oven 1 while the investment and maintenance is only 160k€ more expensive. Even with similar repair and storage costs Oven 3 would be the better choice. (*Table 5*).

Assumptions

Costs related to the oven investment are divided into four main categories: *Investment and Maintenance, Energy Costs, Storage Costs*, and *Repair Costs.*

Investment and Maintenance costs are considered fixed due to the stable pricing of both ovens and maintenance services, as confirmed by Markus Scheer from the purchasing department. These are detailed in *table 2.*

Energy Costs are uncertain based on comments from Scheer and Hadgehorn, thus, electricity and gas prices are simulated for the NPV calculations. Base rates for energy usage and savings are derived from Scheer's quoted suggestions. (See *Table 2*). For details on Energy prices see *table 4*. It assumed that the Ovens are operational 16h a day based on the department running 2 shifts daily.

Storage Costs are calculated by simulating yearly batch sizes based on probabilities in *table 3* Provided by Dorothea Walther of the painting department. If batch sizes exceed oven capacity, a daily storage cost is added (*table 4*). It is assumed that Storage accumulates, also the daily batch is stored incase of a Oven failure. based on Walther's mention of the ability to reschedule production overnight.

Repair Costs are calculated by simulating daily failures based on probabilities on *table 2* provided by Walther. Failure cost and it's variability is based on Walther's quoted values; 5000€ and 50% respectively.

Model

For each oven, we do the following: First, we randomly generate the batch sizes and daily oven failures for the whole five-year period, based on the probabilities provided. Then, we calculate how many units have to be stored due to insufficient oven capacity or minor failures. The produced units can accumulate in storage for many days, before the oven can process them.

We randomly generate daily gas and electricity prices, using the estimates given by Scheer. These are used to calculate daily savings (for the building) and oven energy costs.

To calculate NPV, all daily cashflows are discounted by a daily discount factor (the daily discount rate is assumed constant such that the yearly discount rate is 5 %), and all yearly cashflows (yearly maintenance) are discounted by the respective yearly discount factor. These are added to the initial CapEx to get the final NPV.

We run the model 12000 times for each oven variant.

The catastrophic failure of the old oven and subsidy for Oven 3 are not considered in the common model. The only effect of the subsidy is obviously that with 10 % chance, the NPV will be higher by 50000€ (i.e. 5000€ on average). The catastrophic failure of the old oven will happen with 27% probability over the 5-year period, which, given the estimate of 500k€ already includes all impacts on production, yields an additional 134k€ cost on average (before discounting).

Regardless of these, the old oven remains the second worst option, and Oven 3 remains the best.

2 Technical appendix

The model is implemented in file `keissi_vektorisoitu.ipynb`. The NPV calculation itself is performed in function `calculate_npv()`, which then generates, based on the failure rates and batch size probabilities, the respective daily failures and batch sizes. The other variables with possible daily fluctuations (such as energy prices) can be optionally randomly drawn in advance (the last axis being the day) and given as parameters to `calculate_npv()`.

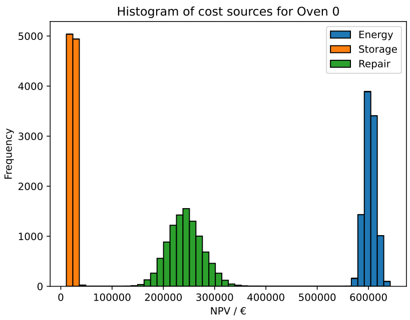
Pretty much everything is vectorised, so the simulation runs in a reasonable time. The only significant Python loop that could not be avoided with basic numpy functionality is the storage simulation (loop over days). However, the code is still fully vectorised over the other dimensions (i.e. all 12000 different scenarios run in parallel w.r.t. the daily loop).

Table 2: Data with technical specifications for the oven investment alternatives

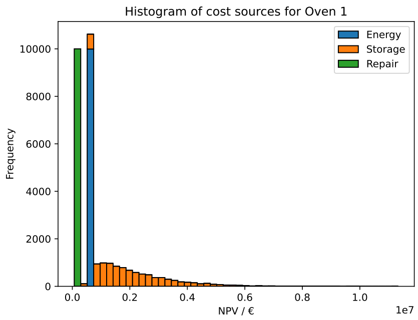
Running Costs	**Current Oven**	**Oven 1**	**Oven 2**	**Oven 3**
Investment cost	€0	€160,000	€240,000	€280,000
Maintenance costs per year	€15,000	€20,000	€25,000	€30,000
Electricity usage (units per hour)	3	2.7	2.5	1.9
Gas usage (units per hour)	2.7	2.2	2.0	1.7
Failure probability (per day)	3%	2%	2%	1%
Capacity (units)	300	180	200	220
Predicted energy usage savings for the whole building				
Gas	0%	5%	10%	15%
Electricity	3%	10%	15%	15%

Table 3: Daily batch sizes and the probability they will break down in the future

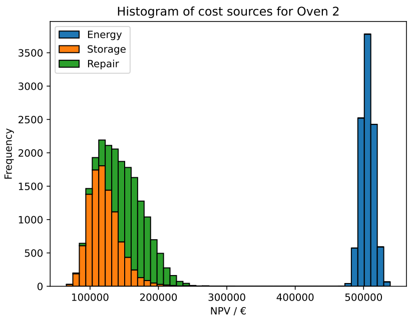
Batch size	**Year 1**	**Year 2**	**Year 3**	**Year 4**	**Year 5**
120	25%	30%	30%	35%	40%
160	25%	25%	30%	30%	30%
200	25%	25%	20%	20%	20%
240	25%	20%	20%	15%	10%



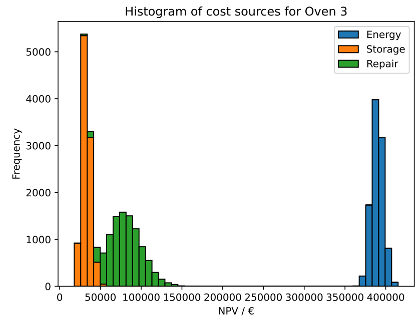
(a) Cost sources Current Oven



(b) Cost sources Oven 1

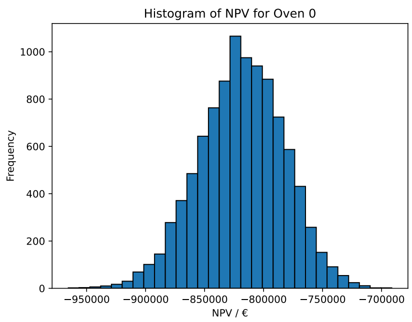


(c) Cost sources Oven 2

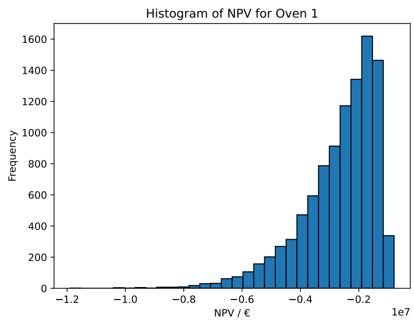


(d) Cost sources Oven 3

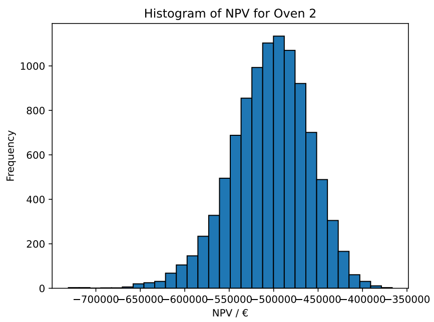
Figure 1: Net Present Value (NPV) cost distribution for four oven investment scenarios.



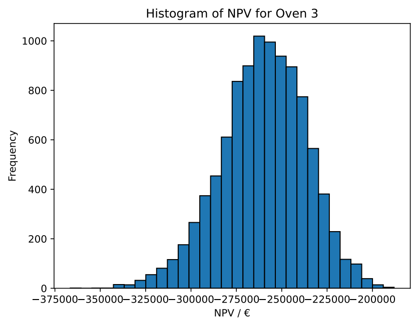
(a) NPV Current Oven



(b) NPV Oven 1



(c) NPV Oven 2



(d) NPV Oven 3

Figure 2: Net Present Value (NPV) simulation results for four oven investment scenarios.

Table 4: Model Parameters and Assumptions

Parameter	**Value / Distribution**
Financial Parameters	
Discount rate	5% annually (applied daily)
Working days per year	365
Horizon	5 years
Baseline Energy Use	
Gas usage	54.6 units/day
Electricity usage	318.0 units/day
Energy Prices (stochastic)	
Gas price	LogNorm($\mu = 1.30$, $\sigma = 0.7$) €/unit
Electricity price	LogNorm($\mu = 1.80$, $\sigma = 0.7$) €/unit
Oven Options	
CAPEX (€)	0, 160k, 240k, 280k
Annual maintenance (€)	15k, 20k, 25k, 30k
Energy Savings	
Gas savings	0%, 5%, 10%, 15%
Electricity savings	0%, 10%, 15%, 15%
Subsidy (stochastic)	
Oven 3 subsidy	€50k with $p = 0.10$ (Year 0), *applied ex post, not in the basic model*
Failure Events (stochastic)	
Normal failures	3%, 2%, 2%, 1% per day
Failure cost	€5,000 per event
Catastrophic failure (current only)	Annual hazard 2%, 4%, 6%, 8%, 10%; €500,000 once
Production and Storage (stochastic)	
Batch size	One batch per day
Capacities	300, 180, 200, 220 units
Batch probabilities	See Table 2 (year-wise)
Storage cost	€2 per excess unit per day
Accounting Note	
Depreciation	Excluded from cash NPV by design

Table 5: Oven Cost distributions

Category	NPV (€)	Standard deviation
Current Oven		
NPV	-818707	35142
Storage costs	23747	3921
Repair costs	242952	32809
Oven energy costs	603224	11279
Building energy savings	119406	2231
Investment and yearly maintenance	68189	0
Oven 1		
NPV	-2664647	1304917
Storage costs	2124124	1295802
Repair costs	161770	26963
Oven energy costs	546583	10137
Building energy savings	418749	7447
Investment and yearly maintenance	250919	0
Oven 2		
NPV	-504079	43411
Storage costs	121334	23407
Repair costs	161770	26963
Oven energy costs	505814	9386
Building energy savings	638488	11182
Investment and yearly maintenance	353649	0
Oven 3		
NPV	-260137	23101
Storage costs	32529	5443
Repair costs	80850	19178
Oven energy costs	389597	7133
Building energy savings	659217	11216
Investment and yearly maintenance	416379	0